

Unit II

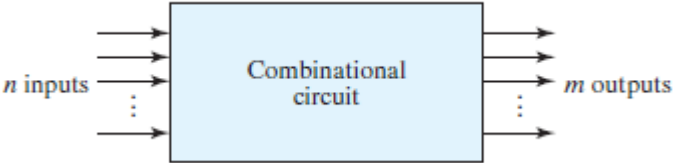
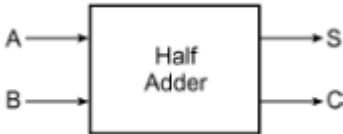
COMBINATIONAL CIRCUITS

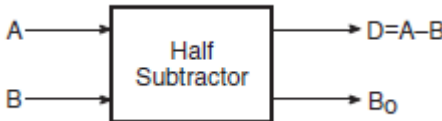
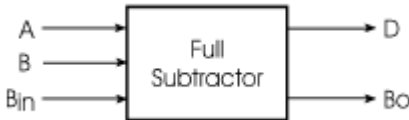
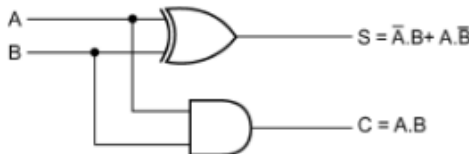
Course Objective – COB1: To analyze and design combinational circuits

Course Outcome - CO1 : Design combinational circuits and analyze the design procedures

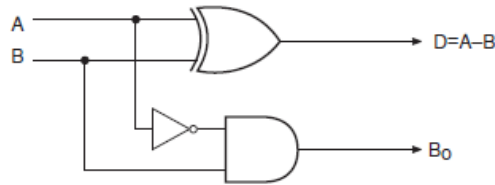
Bloom's Taxonomy Level: K1,K2,K3

Part A

S.No.	Question	BTL*
1.	Define Combinational Circuit. • A combinational circuit consists of logic gates whose outputs at any time are determined from only the present combination of inputs. • A combinational circuit performs an operation that can be specified logically by a set of Boolean functions. 	K1
2.	Define half adder and full adder. Half Adder: A half-adder is an arithmetic circuit block that can be used to add two bits and produce two outputs SUM and CARRY.  Full Adder: A Full-adder is an arithmetic circuit block that can be used to add three bits and produce two outputs SUM and CARRY. 	K1

3.	Define half subtractor and full subtractor. Half subtractor: It is a combinational circuit that subtracts two bits and produces their difference and borrow.  Full subtractor: It is a combinational circuit that performs a subtraction between 2 bits. It also takes into account borrow of the lower significant stage. 	K1																				
4.	Write down the truth table of half adder with logic diagram. Truth Table: <table><tr><th>A</th><th>B</th><th>S</th><th>C</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr></table> Logic Diagram:  The Boolean expressions for the SUM and CARRY outputs are given by the equations $\text{SUM } S = A.\bar{B} + \bar{A}.B$ $\text{CARRY } C = A.B$	A	B	S	C	0	0	0	0	0	1	1	0	1	0	1	0	1	1	0	1	K1
A	B	S	C																			
0	0	0	0																			
0	1	1	0																			
1	0	1	0																			
1	1	0	1																			
5.	Write down the truth table of half subtractor with logic diagram.	K1																				

A	B	D	B ₀
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



Boolean expression for difference and borrow is:

$$D = \bar{A}.B + A.\bar{B}$$

$$B_0 = \bar{A}.B$$

6. What is BCD Adder?

K1

- A BCD(binary-coded decimal)Adder Circuit that adds two BCD digits and produces a sum digit also in BCD.
- BCD numbers use 10 digits, 0 to 9 which are represented in the binary form 0 0 0 0 to 1 0 0 1, i.e. each BCD digit is represented as a 4-bit binary number.

Example:

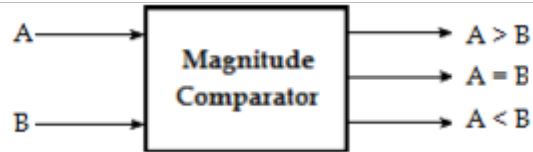
6	0 1 1 0	← BCD for 6
+ 8	1 0 0 0	← BCD for 8
14	1 1 1 0	← Invalid BCD number
	+ 0 1 1 0	← Add 6 for correction
0 0 0 1	0 1 0 0	
		← BCD for 14

7. What is Magnitude Comparator?

K1

- It is a combinational circuit that compares two numbers and determines their relative magnitude.
- The output of comparator is usually 3 binary variables indicating:

$$A > B, A = B, A < B$$



8. Write the truth table of 1 bit comparator with K-map result.

K1

A	B	A>B	A=B	A<B
0	0	0	1	0
1	0	1	0	0
0	1	0	0	1
1	1	0	1	0

		A>B	
A	B	0	1
	0	0	0
1	1	1	0

Equation is $A > B = A \cdot \bar{B}$

		A<B	
A	B	0	1
	0	0	1
1	1	0	0

Equation is $A < B = \bar{A} \cdot B$

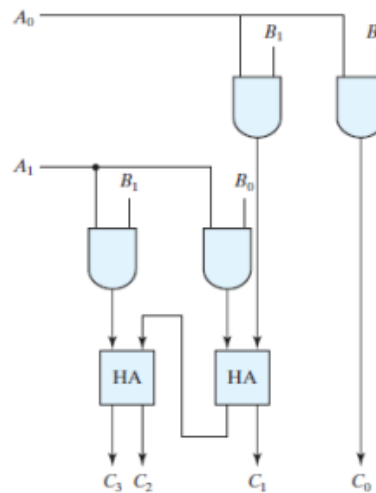
		(A=B)	
A	B	0	1
	0	1	0
1	1	0	1

The equation is $f(A=B) = \bar{A} \cdot \bar{B} + A \cdot B$
 $= A \text{ XNOR } B$

9. Write the working of 2*2 binary multiplier with logic diagram.

K1

	B_1	B_0
A_1	$A_1 B_1$	$A_1 B_0$
A_0	$A_0 B_1$	$A_0 B_0$
	C_3	C_2
	C_1	C_0



10.	What is decoder? • A decoder is a combinational circuit that converts binary information from n input lines to a maximum of 2^n unique output lines. • If the n -bit coded information has unused combinations, the decoder may have fewer than 2^n outputs. • The purpose of a decoder is to generate the 2^n (or fewer) minterms of n input variables	K1																																										
11.	Draw the truth table and logic diagram of 2 to 4 decoder. <table><tr><th>E</th><th>A</th><th>B</th><th>D₀</th><th>D₁</th><th>D₂</th><th>D₃</th></tr><tr><td>1</td><td>X</td><td>X</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> (b) Truth table (a) Logic diagram	E	A	B	D ₀	D ₁	D ₂	D ₃	1	X	X	1	1	1	1	0	0	0	0	1	1	1	0	0	1	1	0	1	1	0	1	0	1	1	0	1	0	1	1	1	1	1	1	K1
E	A	B	D ₀	D ₁	D ₂	D ₃																																						
1	X	X	1	1	1	1																																						
0	0	0	0	1	1	1																																						
0	0	1	1	0	1	1																																						
0	1	0	1	1	0	1																																						
0	1	1	1	1	1	1																																						
12.	What is an encoder? • An encoder is a digital circuit that performs the inverse operation of a decoder. • An encoder has 2^n (or fewer) input lines and n output lines. • The output lines, as an aggregate, generate the binary code corresponding to the input value.	K1																																										
13.	What is multiplexer? • A multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line. • The selection of a particular input line is controlled by a set of selection lines. • Normally, there are 2n input lines and n selection lines whose bit combinations determine which input is selected.	K1																																										
14.	What is 2 to 1 MUX?	K1																																										

- The 2×1 multiplexer is basic two input multiplexer which has two data input lines designated as I_0 and I_1 , one data select line denoted by S and one output line denoted by Y .
- The 2×1 mux is used to connect two 1-bit data sources to a common designation.

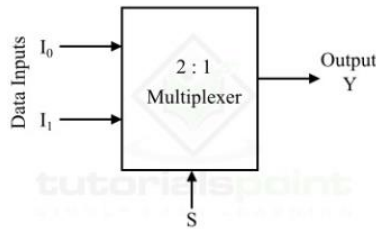
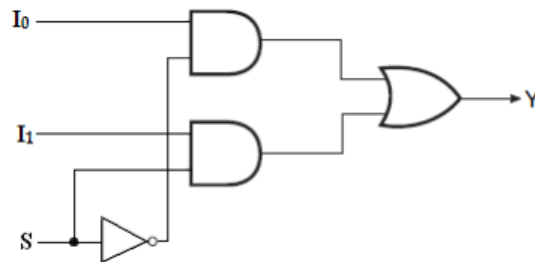


Figure 2 - 2:1 Multiplexer

15. Write the truth table and logic gate of 2 : 1 MUX.

K1

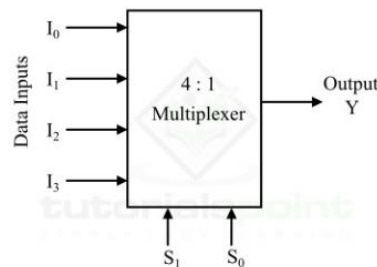
S	Y
0	I_0
1	I_1



16. What is 4 to 1 MUX?

K1

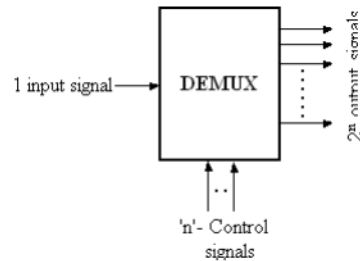
- 4×1 Multiplexer has four data inputs I_3 , I_2 , I_1 & I_0 , two selection lines s_1 & s_0 and one output Y .
- One of these 4 inputs will be connected to the output based on the combination of inputs present at these two selection lines.



17. What is demultiplexer?

K1

- The de-multiplexer performs the inverse function of a multiplexer, that is it receives information on one line and transmits its onto one of 2^n possible output lines.
- The selection is by n input select lines.



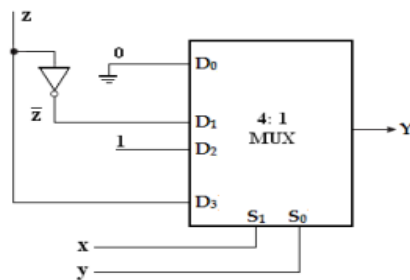
18. Implement the boolean function using Multiplexer. $F(x, y, z) = \sum m(1, 2, 6, 7)$

K3

Solution:
Implementation table:

	D ₀	D ₁	D ₂	D ₃
$\bar{z}$	0	1	2	3
z	4	5	6	7
	0	$\bar{z}$	1	z

Multiplexer Implementation:

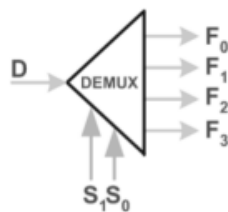


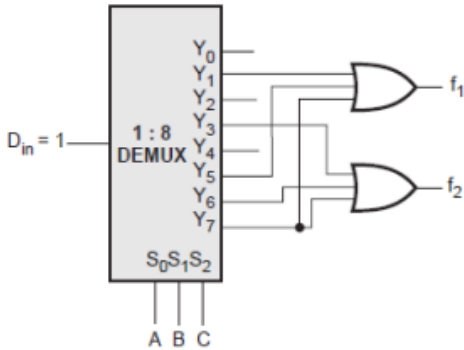
19. Write the truth table of 1 to 4 demultiplexer.

K1

Truth table

S1	S0	F0	F1	F2	F3
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D



20.	Implement the following functions using de-multiplexer. $f_1(A,B,C) = \sum m(1,5,7)$, $f_2(A,B,C) = \sum m(3,6,7)$ Solution: 	K3
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Part B

S.No.	Question	BTL *
1.	(a) Construct a half adder with necessary diagrams. (8) (b) Construct a half subtractor with necessary diagrams (8)	K3
2.	Design a full adder using necessary diagram. How to construct a full adder using 2 half adders. Draw the logic diagram. (16)	K3
3.	Design a full subtractor using necessary diagram. How to construct a full subtractor using 2 half subtractors. Draw the logic diagram. (16)	K3
4.	What is BCD adder? Design an adder to perform arithmetic addition of two decimal bits in BCD (16)	K3
5.	Explain binary multiplication & Draw circuit diagram of 2*2 and 3*4 binary multiplier. (16)	K3
6.	Design 1 bit and 2 bit magnitude comparator with necessary diagrams (16)	K3
7.	Construct 2 to 4 & 3 to 8 decoders with necessary diagrams. How to build 3 to 8 decoder with 2 to 4 decoder. Draw the circuit diagram. (16)	K3
8.	(a) Construct 4:1 MUX with circuit, truth table and logic diagram (8) (b) Implement full adder using De-multiplexer. (8)	K3